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Mechanical interlock system for a filter

Abstract

A mechanical interlock system for a fluid filter is described herein. The mechanical interlock system may include a filter cartridge and a filter cartridge manifold for receiving the filter cartridge. The filter cartridge may include a filter body, a fluid inlet and outlet, and an integrated locking head. The integrated locking head may include a key extending radially outward from the integrated locking head and channels on an outer surface of the integrated locking head. The filter cartridge manifold may include an insertion and extraction tube for receiving the key and latching mechanisms. The insertion and extraction tube may include a primary keyway and a secondary keyway. The key, keyways, latching mechanisms, and/or channels may interact to maintain alignment of the filter cartridge during insertion of the filter cartridge into an appliance including the filter cartridge manifold.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application claims the benefit of U.S. patent application Ser. No. 17/333,124 for a “Mechanical Interlock System for a Filter” filed May 28, 2021 (and published Dec. 1, 2022, as U.S. Patent Application Publication No. 2022/0379248), now U.S. Pat. No. 11,779,867. Each of the foregoing patent application, publication, and patent is hereby incorporated by reference herein.

FIELD OF THE INVENTION

(2) The present invention embraces a mechanical interlock system for a fluid filter, such as a water filter in an appliance (e.g., a refrigerator, a freezer, and/or the like).

BACKGROUND

(3) Appliances, such as refrigerators, freezers, and/or the like, may include one or more components, such as a water dispenser, an ice maker, and/or the like, which use water from a water supply (e.g., a household water supply and/or the like). Such appliances typically include a water filter that filters water received from the water supply before the water is provided to the one or more components. Such water filters have a useful lifespan that is shorter than the appliance and must be replaced for the appliance to continue to provide properly filtered water. Appliances typically include an opening to permit a user to remove an expired water filter and install a replacement water filter.

SUMMARY

(4) The following presents a simplified summary of one or more embodiments of the present invention, in order to provide a basic understanding of such embodiments. This summary is not an extensive overview of all contemplated embodiments and is intended to neither identify key or critical elements of all embodiments nor delineate the scope of any or all embodiments. This summary presents some concepts of one or more embodiments of the present invention in a simplified form as a prelude to the more detailed description that is presented later.

(5) In one aspect, the present invention embraces a mechanical interlock system, which may include a filter cartridge and a filter cartridge manifold for receiving the filter cartridge. The filter cartridge may include a filter body having a first end, a second end, and a longitudinal axis, a fluid inlet and outlet at the first end of the filter body, and an integrated locking head on the first end of the filter body. The integrated locking head may include a key extending radially outward from the integrated locking head and a channel on an outer surface of the integrated locking head structured for receiving a latching mechanism, preventing rotation of the filter cartridge in a first direction, and permitting rotation of the filter cartridge in a second direction. The filter cartridge manifold may include an insertion and extraction tube for receiving the key and the latching mechanism. The insertion and extraction tube may include a primary keyway and a secondary keyway, where the secondary keyway has an entry end and an exit end, and where the exit end of the secondary keyway connects to the primary keyway. The latching mechanism may be structured to prevent, after being received by the channel, movement of the filter cartridge in a direction along the longitudinal axis that is opposite an insertion direction, prevent rotation of the filter cartridge in the first direction, and permit rotation of the filter cartridge in the second direction.

(6) In some embodiments, during insertion of the filter cartridge into the filter cartridge manifold, the key passes through the primary keyway.

(7) In some embodiments, when the key is positioned in the primary keyway, the primary keyway prevents rotation of the filter cartridge, aligns the latching mechanism with the channel, and aligns the fluid inlet and outlet of the filter cartridge with a fluid connector of the filter cartridge manifold, and wherein the filter cartridge manifold is positioned in an appliance.

(8) In some embodiments, an end portion of the primary keyway includes a sidewall configured to interact with the key to prevent rotation of the filter cartridge in a first direction, and the end portion permits rotation of the filter cartridge in a second direction. Additionally, or alternatively, after insertion of the filter cartridge into the filter cartridge manifold, a vertical sidewall of the

latching mechanism interacts with a vertical sidewall of the channel to prevent rotation of the filter cartridge in the first direction. In some embodiments, after insertion of the filter cartridge into the filter cartridge manifold, a filleted sidewall of the latching mechanism interacts with a filleted sidewall of the channel to permit rotation of the filter cartridge in the second direction.

Additionally, or alternatively, after insertion of the filter cartridge into the filter cartridge manifold and during rotation of the filter cartridge in the second direction the filleted sidewall of the latching mechanism interacts with the filleted sidewall of the channel to force the latching mechanism to flex radially outward and slide over an outer circumference of the integrated locking head and the key rotates toward the entry end of the secondary keyway.

(9) In some embodiments, the filter cartridge manifold may include a spring element and a pusher component spring-loaded by the spring element and configured to provide a compressive force against the integrated locking head.

(10) In another aspect, the present invention embraces a filter cartridge. The filter cartridge may include a filter body having a first end, a second end, and a longitudinal axis, a fluid inlet and outlet at the first end of the filter body, and an integrated locking head on the first end of the filter body. The integrated locking head may include a channel on an outer surface of the integrated locking head, where the channel is configured to receive a latching mechanism of a filter cartridge manifold, and where the channel includes a filleted sidewall at a first end of the channel and a vertical sidewall at a second end of the channel.

(11) In some embodiments, the channel may include a chamfered edge oriented towards the fluid inlet and outlet, where insertion of the filter cartridge into the filter cartridge manifold causes the chamfered edge of the channel to interact with a chamfered inner edge of the latching mechanism forcing the latching mechanism to flex radially outward. Additionally, or alternatively, after insertion of the filter cartridge into the filter cartridge manifold, the chamfered inner edge of the latching mechanism is structured to sit within the channel and prevent movement of the filter cartridge in a direction along the longitudinal axis that is opposite an insertion direction.

(12) In some embodiments, after insertion of the filter cartridge into the filter cartridge manifold, a vertical sidewall of the latching mechanism is structured to interact with the vertical sidewall of the channel to prevent rotation of the filter cartridge in a first direction.

(13) In some embodiments, after insertion of the filter cartridge into the filter cartridge manifold, a filleted sidewall of the latching mechanism is structured to interact with the filleted sidewall of the channel to permit rotation of the filter cartridge in a second direction. Additionally, or alternatively, after insertion of the filter cartridge into the filter cartridge manifold and during rotation of the filter cartridge in the second direction, the filleted sidewall of the latching mechanism is structured to interact with the filleted sidewall of the channel to force the latching mechanism to flex radially outward and slide over an outer circumference of the integrated locking head.

(14) In some embodiments, the channel is a first channel, and the integrated locking head includes a second channel on the outer surface of the integrated locking head, where the first channel and the second channel are diametrically opposed on the outer surface of the integrated locking head.

(15) In some embodiments, the integrated locking head may include a key extending radially outward from the integrated locking head. Additionally, or alternatively, during insertion of the filter cartridge into a filter cartridge manifold, the key is structured to pass through a primary keyway of an insertion and extraction tube of the filter cartridge manifold. In some embodiments, when the key is positioned in a primary keyway of an insertion and extraction tube of a filter cartridge manifold, the primary keyway is structured to prevent rotation of the filter cartridge. Additionally, or alternatively, the key may have a rhomboid-shape and a radially-outward-facing surface that is sloped radially inward from a center of the radially-outward-facing surface.

(16) In yet another aspect, the present invention embraces a method. The method may include providing a filter cartridge including an integrated locking head, where the integrated locking head includes a channel on an outer surface of the integrated locking head, and where the channel is

structured to receive a latching mechanism. The method may include, while inserting, engaging the integrated locking head with the latching mechanism forcing the latching mechanism to flex radially outward. The method may include, while inserting and after forcing the latching mechanism to flex radially outward, positioning the latching mechanism in the channel.

(17) In some embodiments, the method may include, while inserting, overcoming a force applied to the integrated locking head in an extraction direction opposite the insertion direction.

(18) In some embodiments, the integrated locking head may include a key extending radially outward from the integrated locking head, and the method may include, before inserting, aligning the key with a keyway of the filter cartridge manifold.

(19) In some embodiments, the method may include, while the latching mechanism is positioned in the channel, preventing movement of the filter cartridge in an extraction direction opposite the insertion direction and preventing, with a vertical sidewall of the channel and a vertical sidewall of the latching mechanism, rotation of the filter cartridge in a first direction. Additionally, or alternatively, the method may include, while the latching mechanism is positioned in the channel, permitting, with a filleted sidewall of the channel and a filleted sidewall of the latching mechanism, rotation of the filter cartridge in a second direction opposite the first direction.

(20) In some embodiments, the method may include, after positioning the latching mechanism in the channel, rotating the filter cartridge in the second direction. Additionally, or alternatively, the method may include, while rotating the filter cartridge in the second direction, forcing, with a filleted sidewall of the channel and a filleted sidewall of the latching mechanism, the latching mechanism to flex radially outward and engage an outer surface of the integrated locking head.

(21) In some embodiments, the method may include, after forcing the latching mechanism to flex radially outward and engage the outer surface of the integrated locking head, pushing the integrated locking head in an extraction direction opposite the insertion direction. Additionally, or alternatively, the integrated locking head may include a key extending radially outward from the integrated locking head, and the method may include, after forcing the latching mechanism to flex radially outward and engage an outer surface of the integrated locking head, pulling the filter cartridge in an extraction direction opposite the insertion direction causing the key to enter a secondary keyway, of the filter cartridge manifold, that intersects with a primary keyway, of the filter cartridge manifold, through which the key passed while inserting the filter cartridge into the filter cartridge manifold.

(22) In some embodiments, the method may include, while pulling the filter cartridge in the extraction direction, pulling the key through the secondary keyway and into the primary keyway and, while pulling the filter cartridge in the extraction direction and pulling the key through the secondary keyway, rotating the filter cartridge in the first direction. Additionally, or alternatively, the method may include, while pulling the filter cartridge in the extraction direction, pulling the key through the primary keyway and withdrawing the filter cartridge from the appliance.

(23) The features, functions, and advantages that have been discussed may be achieved independently in various embodiments of the present invention or may be combined with yet other embodiments, further details of which may be seen with reference to the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Having thus described embodiments of the invention in general terms, reference will now be made to the accompanying drawings, wherein:

(2) FIG. 1 illustrates a perspective view of a filter cartridge, in accordance with an embodiment of the invention;

- (3) FIG. 2 illustrates a close-up, side view of an integrated locking head of the filter cartridge, in accordance with an embodiment of the invention;
- (4) FIG. 3 illustrates another close-up, side view of the integrated locking head of the filter cartridge, in accordance with an embodiment of the invention;
- (5) FIG. 4 illustrates a close-up, side view of the filter cartridge and a portion of a filter cartridge manifold of an appliance, in accordance with an embodiment of the invention;
- (6) FIG. 5 illustrates a perspective view of a portion of the filter cartridge manifold, in accordance with an embodiment of the invention;
- (7) FIG. 6 illustrates a perspective view of the filter cartridge manifold with the filter cartridge inserted therein, in accordance with an embodiment of the invention;
- (8) FIG. 7 illustrates a perspective view of the filter cartridge manifold with the filter cartridge inserted therein with portions of the filter cartridge manifold removed, in accordance with an embodiment of the invention;
- (9) FIG. 8 illustrates another perspective view of the filter cartridge manifold with the filter cartridge inserted therein with portions of the filter cartridge manifold removed, in accordance with an embodiment of the invention;
- (10) FIG. 9 illustrates a side, cross-section view of the integrated locking head of the filter cartridge and latching mechanisms of the filter cartridge manifold after insertion of the filter cartridge into the filter cartridge manifold, in accordance with an embodiment of the invention;
- (11) FIG. 10 illustrates a side, cross-section view of the integrated locking head of the filter cartridge and the latching mechanisms of the filter cartridge manifold after partial rotation of the filter cartridge within the filter cartridge manifold, in accordance with an embodiment of the invention;
- (12) FIG. 11 illustrates a side, cross-section view of the integrated locking head of the filter cartridge and the latching mechanisms of the filter cartridge manifold after further rotation of the filter cartridge within the filter cartridge manifold, in accordance with an embodiment of the invention;
- (13) FIG. 12 illustrates a perspective view of the filter cartridge and the filter cartridge manifold with portions of the filter cartridge manifold removed after rotation of the filter cartridge within the filter cartridge manifold, in accordance with an embodiment of the invention; and
- (14) FIG. 13 illustrates a perspective view of the filter cartridge and the filter cartridge manifold with portions of the filter cartridge manifold removed during removal of the filter cartridge, in accordance with an embodiment of the invention.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

(15) Embodiments of the present invention will now be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all, embodiments of the invention are shown. Indeed, the invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Where possible, any terms expressed in the singular form herein are meant to also include the plural form and vice versa, unless explicitly stated otherwise. Also, as used herein, the term “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein. Furthermore, when it is said herein that something is “based on” something else, it may be based on one or more other things as well. In other words, unless expressly indicated otherwise, as used herein “based on” means “based at least in part on” or “based at least partially on.” Like numbers refer to like elements throughout. Where possible, any of the advantages, features, functions, devices, and/or operational aspects of any of the embodiments of the present invention described and/or contemplated herein may be included in any of the other embodiments of the present invention described and/or contemplated herein, and/or vice versa.

- (16) As used herein, “operatively coupled” may mean that the components may be formed

integrally with each other or may be formed separately and coupled together. Furthermore, “operatively coupled” may mean that the components may be formed directly to each other or to each other with one or more components located between the components that are operatively coupled together. Additionally, or alternatively, “operatively coupled” may mean that the components are detachable from each other or that they are permanently coupled together. Furthermore, “operatively coupled” components may mean that the components retain at least some freedom of movement in one or more directions or may be rotated about an axis (i.e., rotationally coupled, pivotally coupled, and/or the like).

(17) As noted, appliances, such as refrigerators, freezers, and/or the like, may include one or more components, such as a water dispenser, an ice maker, and/or the like, which use water from a water supply (e.g., a household water supply and/or the like). Such appliances typically include a water filter that filters water received from the water supply before the water is provided to the one or more components. Such water filters have a useful lifespan that is shorter than the appliance and must be replaced for the appliance to continue to provide properly filtered water. Appliances typically include an opening to permit a user to remove an expired water filter and install a replacement water filter. However, the user may have difficulty determining the correct steps to remove the expired water filter. Furthermore, the user may have difficulty installing the replacement water filter due to a failure to properly align the replacement water filter with a manifold within the appliance, due to a lack of indication that the replacement water filter has been properly installed, and/or the like.

(18) Some embodiments described herein provide a mechanical interlock system for a filter, such as a water filter, which may be included in an appliance. In some embodiments, the mechanical interlock system may include a filter cartridge and a filter cartridge manifold for an appliance push-to-lock mechanism for securing the filter cartridge and a twist-to-unlock mechanism for releasing the filter cartridge. The filter cartridge may have an integrated locking head, which may include a key extending radially outward from the integrated locking head and one or more channels on an outer surface of the integrated locking head. The mechanical interlock system may include a filter cartridge manifold for receiving the filter cartridge, where the filter cartridge manifold includes an insertion and extraction tube and one or more latching mechanisms (e.g., claws). In some embodiments, the key of the integrated locking head may interact with the insertion and extraction tube to properly align the filter cartridge during installation of the filter cartridge and removal of the filter cartridge. Additionally, or alternatively, the one or more latching mechanisms may interact with the one or more channels on the integrated locking head to permit insertion of the filter cartridge, provide an indication that the filter cartridge has been properly installed, prevent rotation of the filter cartridge in an incorrect direction, and permit rotation of the filter cartridge in a correct direction. In some embodiments, the mechanical interlock system may include a stop element to prevent over-rotation of the filter cartridge in the correct direction.

(19) FIG. 1 illustrates a perspective view of a filter cartridge **10**, in accordance with an embodiment of the invention. As shown in FIG. 1, the filter cartridge **10** may include a filter body **12**, also referred to as a filter housing, which typically encloses a fluid filtration material. The filter body **12** may be substantially cylindrical and may define a first end and a second end along a longitudinal axis. Additionally, or alternatively, the filter body **12** may be slightly conical such that the first end is has a smaller radius than the radius of the second end.

(20) The filter cartridge **10** may include a fluid inlet and outlet **14** as well as an integrated locking head **20**, which may be formed into the filter body **12** and/or attached to the filter body **12** as shown in FIG. 1. In some embodiments, the fluid inlet and outlet **14** may be formed in a central portion of the first end of the filter body **12**, and the integrated locking head may be formed into the filter body **12** and/or attached to the filter body **12** at a peripheral portion of the first end of the filter body **12**. In use, water may pass through the fluid inlet into the filter body **12**, be filtered by the fluid filtration material, and pass through the fluid outlet to the appliance.

(21) As shown in FIG. 1, the integrated locking head **20** may include a key **22** and a channel **24**. In some embodiments, the key **22** may extend radially outward from the integrated locking head **20** (e.g., from an outer surface of the integrated locking head **20** and/or the like). Although not shown in FIG. 1, the integrated locking head **20** may include another channel positioned on the opposite side of the integrated locking head **20** from the channel **24**. In some embodiments, the channels **24** may be positioned and/or formed on an outer surface of the integrated locking head **20**.

Additionally, or alternatively, the channels **24** may be diametrically opposed on the outer surface of the integrated locking head. In some embodiments, and as described herein, the channels **24** may be structured and/or configured for receiving latching mechanisms, preventing rotation of the filter cartridge **10** in a first direction, and permitting rotation of the filter cartridge **10** in a second direction.

(22) FIG. 2 illustrates a close-up, side view of the integrated locking head **20** of the filter cartridge **10**. FIG. 3 illustrates another close-up, side view of the integrated locking head **20** of the filter cartridge **10**. As shown in FIG. 2, the channel **24** may include a chamfered edge **24a** oriented towards the fluid inlet and outlet **14**, a vertical sidewall **24b** on a first end of the channel **24**, and a filleted or curved sidewall **24c** on a second end of the channel **24**. As described further herein, the chamfered edge **24a**, the vertical sidewall **24b**, and the filleted sidewall **24c** may interact with corresponding features on a lock mechanism on a filter cartridge manifold. As shown in FIG. 3, the key **22** may be shaped as an extruded rhombus with a tip that is sloped from its center, and, as described further herein, the key **22** may interact with keyways in a tube of the filter cartridge manifold. In some embodiments, the key **22** may have a rhomboid-shape and a radially-outward-facing surface that is sloped radially inward from a center of the radially-outward-facing surface.

(23) FIG. 4 illustrates a close-up, side view of the filter cartridge and a portion of a filter cartridge manifold of an appliance. FIG. 5 illustrates a perspective view of a portion of the filter cartridge manifold. As shown in FIG. 4, the filter cartridge manifold may include a first base component **26a**, a locking mechanism including two latching mechanisms **28**, and a fluid connector **16**. In some embodiments, the first base component **26a** may be connected to the locking mechanism and/or the latching mechanisms **28**. Additionally, or alternatively, the first base component **26a** may be connected to the appliance (e.g., via fasteners and/or the like) and/or may be connected to a second base component **26b** (See FIG. 6), which may be connected to the appliance. The first base component **26a** and/or the second base component **26b** may maintain positioning of the locking mechanism including the latching mechanisms **28** within the appliance, with respect to the filter cartridge manifold, and with respect to the fluid connector **16** in the appliance. The fluid connector **16** of the filter cartridge manifold provides a fluid connection between the internal components of the appliance and the fluid inlet and outlet **14** of the filter cartridge **10**.

(24) As shown in FIGS. 4 and 5, the filter cartridge manifold may also include a pusher component **30** that is spring-loaded by a spring element **32** (See FIG. 5). In some embodiments, the spring element **32** is positioned between the pusher component **30** and the first base component **26a** and/or the second base component **26b**. Additionally, or alternatively, during installation of the filter cartridge **10**, the pusher component **30** may provide a compressive force against the integrated locking head **20** that assists with seating the two latching mechanisms **28** in the corresponding channels **24** on opposite sides of the integrated locking head **20** (See FIG. 4).

(25) As shown in FIG. 5, the latching mechanisms **28** may include a chamfered inner edge **28a**, a vertical sidewall **28b**, and a filleted or curved sidewall **28c**. The chamfered inner edge **28a** may interact with the chamfered edge **24a** of the integrated locking head **20** shown in FIGS. 2 and 3 such that, as the filter cartridge **10** is inserted into the filter cartridge manifold, the latching mechanisms **28** flex radially outward allowing the integrated locking head **20** to insert into the filter cartridge manifold and/or the fluid inlet and outlet **14** to connect to a corresponding fluid connector **16** on the appliance. After the chamfered inner edges **28a** pass into the channels **24** of the locking head **20**, the latching mechanisms **28** may sit within the channels **24** (See FIG. 4) and lock the filter

cartridge **10** within the filter cartridge manifold by preventing movement in a direction along the longitudinal axis that is opposite the insertion direction. The seating of the latching mechanisms **28** within the channels **24** may generate a noise indicating that the filter cartridge **10** is locked in the filter cartridge manifold.

(26) As shown in FIGS. **4** and **5**, the latching mechanisms **28** may include a neck portion **28d** extending parallel to the longitudinal axis in a direction opposite the insertion direction and a head portion **28e** at an end of the neck portion **28d**. In some embodiments, the head portion **28e** may have a curvature complementary to an outer curvature of the integrated locking head **20**.

Additionally, or alternatively, the head portion **28e** may be angled inward from an outer edge to an inner edge, which may guide the integrated locking head **20** into a space between the latching mechanisms **28** during insertion.

(27) In some embodiments, and as shown in FIGS. **4** and **5**, the outer edge of the head portion may extend radially outward beyond the neck portion **28d**, which may improve an ability of the latching mechanisms **28** to guide the integrated locking head **20** into a space between the latching mechanisms **28** during insertion. Additionally, or alternatively, and as also shown in FIGS. **4** **5**, the inner edge of the head portion **28e** may extend radially inward from the neck portion **28d**, which may permit the latching mechanisms **28** to sit in the channels **24** and prevent movement of the filter cartridge **10** in a direction opposite the insertion direction. In some embodiments, and as also shown in FIGS. **4** and **5**, an angle α between the inner edge of the head portion and the neck portion may be less than 90 degrees, which may improve an ability of the latching mechanisms **28** to prevent movement of the filter cartridge in a direction opposite the insertion direction. In some embodiments, and as described herein, the latching mechanisms **28** may be structured and/or configured to prevent, after being received by the channels **24**, movement of the filter cartridge **10** in a direction along the longitudinal axis that is opposite an insertion direction, prevent rotation of the filter cartridge in a first direction, and permit rotation of the filter cartridge **10** in a second direction.

(28) FIG. **6** illustrates a perspective view of the filter cartridge manifold **40** with the filter cartridge inserted therein. FIG. **7** illustrates a perspective view of the filter cartridge manifold with the filter cartridge inserted therein with portions of the filter cartridge manifold removed. As shown in FIGS. **6** and **7**, the filter cartridge manifold **40** may include an insertion and extraction tube **42** having two keyways **34** and **36** (e.g., a primary keyway **34** and a secondary keyway **36**) that intersect. In some embodiments, keyway **34** may be straight and extend parallel to the longitudinal axis of the filter cartridge **10**. When a user inserts the filter cartridge **10**, the key **22** on the integrated locking head **20** may pass through keyway **34**, and keyway **34** may prevent rotation of the filter cartridge **10** during insertion.

(29) In some embodiments, when the key **22** is positioned in the keyway **34**, the keyway **34** prevents rotation of the filter cartridge **10**, aligns one of the two latching mechanisms **28** with one of the two channels **24** and another of the two latching mechanisms **28** with another of the two channels **24**, and aligns the fluid inlet and outlet **14** of the filter cartridge **10** with the fluid connector **16** of an appliance. In some embodiments, and as shown in FIGS. **6** and **7**, the insertion and extraction tube **42** may include an insertion end **34a** for receiving and guiding the key **22** into the keyway **34**, where the insertion end **34a** is wider than the keyway **34**, which may improve the ability of a user to insert the key **22** into the keyway **34** and the filter cartridge **10** into the appliance.

(30) As shown in FIGS. **6** and **7**, keyway **36** may be curved and may include a wedge element **36a** and a leaf spring element **36b**. The wedge element **36a** may be configured to guide the key **22** into the keyway **36** after the pusher component **30** pushes the filter cartridge **10**. In some embodiments, keyway **36** may have an entry end and an exit end, where the exit end of keyway **36** connects to keyway **34**. The leaf spring element **36b**, as shown in FIG. **6**, may be biased downward to prevent the key **22** from entering keyway **36** during insertion and to flex upward to permit the key **22** to

exit keyway **36** and enter keyway **34** during extraction of the filter cartridge **10**. Although the leaf spring element **36b** shown in FIG. **6** is biased downward into keyway **36**, the leaf spring element **36b** may be biased into the keyway **36** from another direction (e.g., from a side of keyway **36** and/or the like). The curved shape of keyway **36** may reduce any inertia the filter cartridge **10** may have during extraction (e.g., due to being pushed by the pusher component **30**).

(31) FIG. **8** illustrates another perspective view of the filter cartridge manifold **40** with the filter cartridge inserted therein with portions of the filter cartridge manifold **40** removed. As shown in FIG. **8**, the end portion of keyway **34** may include an end portion with a sidewall **34b** and/or the like that interacts with the key **22** to prevent clockwise rotation (e.g., from a user's perspective) of the filter cartridge **10** but permits counterclockwise rotation of the key **22** and the filter cartridge **10**. As also shown in FIG. **8**, the filter cartridge manifold **40** may include a stop element **36c** that interacts with the key **22** to prevent over-rotation of the filter cartridge **10** in a counterclockwise direction. In some embodiments, the stop element **36c** have a curved surface to guide the key **22** into keyway **36**.

(32) FIG. **9** illustrates a side, cross-section view of the integrated locking head **20** of the filter cartridge and latching mechanisms **28** of the filter cartridge manifold after insertion of the filter cartridge into the filter cartridge manifold. As shown in FIG. **9**, the inner radius of the latching mechanisms **28** is less than the outer radius of the integrated locking head **20**, such that the latching mechanisms **28**, when seated in the channels **24**, prevent movement of the filter cartridge **10** in a direction opposite the insertion direction. As also shown in FIG. **9**, the vertical sidewalls **28b** of the latching mechanisms **28** may interact with the vertical sidewalls **24b** of the channels **24** in the integrated locking head **20** to prevent clockwise rotation (e.g., from a user's perspective) of the filter cartridge **10**.

(33) FIG. **10** illustrates a side, cross-section view of the integrated locking head **20** of the filter cartridge and the latching mechanisms **28** of the filter cartridge manifold after partial rotation of the filter cartridge within the filter cartridge manifold. As shown in FIG. **10**, the filleted sidewalls **28c** may interact with the filleted sidewalls **24c** of the channels **24** to permit counterclockwise rotation (e.g., from the user's perspective) of the filter cartridge **10**.

(34) FIG. **11** illustrates a side, cross-section view of the integrated locking head **20** of the filter cartridge and the latching mechanisms **28** of the filter cartridge manifold after further rotation of the filter cartridge within the filter cartridge manifold. As shown in FIG. **10**, the filleted sidewalls **28c** may interact with the filleted sidewalls **24c** of the channels **24** to force, after sufficient counterclockwise rotation (e.g., from the user's perspective) of the filter cartridge **10**, the two latching mechanisms **28** to flex radially outward and slide over the outer circumference of the integrated locking head **20**, thereby unlocking the integrated locking head **20** from the filter cartridge manifold.

(35) FIG. **12** illustrates a perspective view of the filter cartridge and the filter cartridge manifold **40** with portions of the filter cartridge manifold **40** removed after rotation of the filter cartridge within the filter cartridge manifold **40**. As shown in FIG. **12**, after a user rotates the filter cartridge in a counterclockwise direction, from the user's perspective, the latching mechanisms **28** may slide out of the channels **24** (e.g., due to the filleted sidewalls **28c** of the latching mechanisms **28** and the filleted sidewalls **24c** of the channels **24** as described with respect to FIGS. **9-11**), flex radially outward, and slide over the outer circumference of the integrated locking head **20**. Furthermore, and as shown in FIG. **12**, after the user rotates the filter cartridge in the counterclockwise direction, the key **22** rotates towards the stop element **36c**. As the filter cartridge becomes unlocked from filter cartridge manifold **40** due to the latching mechanisms **28** sliding out of the channels **24**, the pusher component **30**, which is spring-loaded by the spring element **32** (See FIG. **5**), may push the filter cartridge **10** in a direction opposite the insertion direction causing the key **22** to enter keyway **36**. As described herein, the wedge element **36a** and/or the stop element **36c** may guide the key **22** into keyway **36**. Furthermore, as the user pulls the filter cartridge in a direction opposite the

insertion direction, the key 22 may pass through keyway 36, and the leaf spring element 36b may flex to permit the key 22 to pass through keyway 36.

(36) FIG. 13 illustrates a perspective view of the filter cartridge and the filter cartridge manifold 40 with portions of the filter cartridge manifold 40 removed during removal of the filter cartridge. As shown in FIG. 13, as the user pulls the filter cartridge in a direction opposite the insertion direction, the key 22 may exit keyway 36 and enter keyway 34. The user may continue to pull the filter cartridge, and the key 22 may pass through keyway 34 toward the insertion end 34a until the filter cartridge is completely removed from the filter cartridge manifold 40 and/or the appliance.

(37) In some embodiments, the present invention embraces a method of inserting a filter cartridge into a filter manifold and/or removing a filter cartridge from a filter manifold. First, the method may include aligning the key 22 (FIGS. 1-3) with the insertion end 34a of keyway 34 of the insertion and extraction tube 42 (FIGS. 6-8) and inserting, in an insertion direction, the filter cartridge 10 into the filter cartridge manifold 40 such that the key 22 passes through keyway 34. While inserting the filter cartridge 10 in the insertion direction, the leaf spring element 36b (FIGS. 6-8) typically prevents the key 22 from entering keyway 36. Next, the method may include engaging the integrated locking head 20 with the latching mechanisms 28 (FIGS. 4-5) and forcing the latching mechanisms 28 to flex radially outward. The method may also include engaging the integrated locking head 20 with the pusher component 30 (FIGS. 4-5) and overcoming a force applied by the pusher component 30 to the integrated locking head 20 in a direction opposite the insertion direction. The method may further include, after forcing the latching mechanisms 28 to flex radially outward, positioning the latching mechanisms 28 in the channels 24 (FIGS. 4 and 7-9). The method may also include applying a force, with the pusher component 30, to the integrated locking head 20 in a direction opposite the insertion direction after the latching mechanisms 28 are positioned in the channels 24 and preventing movement of the filter cartridge 10 in the direction opposite the insertion direction. In some embodiments, the method may include, after the latching mechanisms 28 are positioned in the channels 24, preventing, with the vertical sidewalls 24b of the channels 24 (FIGS. 2 and 9-11) and the vertical sidewalls 28b of the latching mechanisms 28 (FIGS. 5 and 9-11), rotation of the filter cartridge 10 in a first direction (e.g., a clockwise direction).

(38) As noted, the present invention may embrace a method of removing a filter cartridge from a filter manifold. The method may include, after the latching mechanisms 28 are positioned in the channels 24, permitting, with the filleted sidewalls 24c of the channels 24 (FIGS. 2 and 9-11) and the filleted sidewalls 28c of the latching mechanisms 28 (FIGS. 5 and 9-11), rotation of the filter cartridge 10 in a second direction opposite the first direction (e.g., a counterclockwise direction). The method may include, after the latching mechanisms 28 are positioned in the channels 24, rotating the filter cartridge 10 in the second direction and forcing, with the filleted sidewalls 24c of the channels 24 and the filleted sidewalls 28c of the latching mechanisms 28, the latching mechanisms 28 to flex radially outward and engage the outer surface of the integrated locking head 20 (FIGS. 11-12).

(39) In some embodiments, the method may include, after forcing the latching mechanisms 28 to flex radially outward and engage the outer surface of the integrated locking head 20, pushing, with the pusher component 30, the integrated locking head 20 and the filter cartridge 10 in an extraction direction opposite the insertion direction, such that the key 22 enters keyway 36. The method may include preventing over-rotation of the filter cartridge 10 with the stop element 36c (FIG. 12). Additionally, or alternatively, the method may include guiding the key 22 into keyway 36 with the wedge element 36a (FIGS. 7-8 and 12-13). The method may also include pulling the filter cartridge 10 in the extraction direction such that the key 22 passes through keyway 36, pushes the leaf spring element 36b upward, and enters keyway 34 (FIG. 13). The method may also include, while pulling the filter cartridge 10 in the extraction direction and while the key 22 passes through keyway 36, rotating the filter cartridge 10 in the first direction (e.g., a clockwise direction). The method may

further include pulling the filter cartridge **10** in the extraction direction such that the key **22** passes through keyway **34** and the insertion end **34a** and withdrawing the filter cartridge **10**.

(40) Although many embodiments of the present invention have just been described above, the present invention may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. Also, it will be understood that, where possible, any of the advantages, features, functions, devices, and/or operational aspects of any of the embodiments of the present invention described and/or contemplated herein may be included in any of the other embodiments of the present invention described and/or contemplated herein, and/or vice versa. In addition, where possible, any terms expressed in the singular form herein are meant to also include the plural form and/or vice versa, unless explicitly stated otherwise. Accordingly, the terms “a” and/or “an” shall mean “one or more,” even though the phrase “one or more” is also used herein. Like numbers refer to like elements throughout.

(41) While certain exemplary embodiments have been described and shown in the accompanying drawings, it is to be understood that such embodiments are merely illustrative of and not restrictive on the broad invention, and that this invention not be limited to the specific constructions and arrangements shown and described, since various other changes, combinations, omissions, modifications and substitutions, in addition to those set forth in the above paragraphs, are possible. Those skilled in the art will appreciate that various adaptations, modifications, and combinations of the just described embodiments may be configured without departing from the scope and spirit of the invention. Therefore, it is to be understood that, within the scope of the appended claims, the invention may be practiced other than as specifically described herein.

Claims

1. A filter cartridge, comprising: a filter body having a first end, a second end, and a longitudinal axis; a fluid inlet and outlet at the first end of the filter body; an integrated locking head on the first end of the filter body, wherein the integrated locking head comprises a first component on an outer surface of the integrated locking head, wherein the first component is configured to engage a second component of a filter cartridge manifold; and a key extending radially outward from the filter body adjacent the first end; wherein one of the first component and the second component is a pair of channels comprising a first channel and a second channel, wherein the first channel and the second channel are diametrically opposed with respect to the longitudinal axis; wherein the other of the first component and the second component is a pair of latching mechanisms comprising a first latching mechanism and a second latching mechanism, and wherein the first latching mechanism is configured to be received by the first channel and the second latching mechanism is configured to be received by the second channel; wherein the first channel and the second channel each comprise a curved wall at an end of each respective channel; and wherein, during insertion of the filter cartridge into the filter cartridge manifold, the key is structured to pass through a primary keyway of the filter cartridge manifold.
2. The filter cartridge of claim 1, wherein the first latching mechanism is configured to, after being received by the first channel, sit within the first channel, and wherein the second latching mechanism is configured to, after being received by the second channel, sit within the second channel.
3. The filter cartridge of claim 1, wherein each of the first latching mechanism and the second latching mechanism comprises a neck portion and an end portion, wherein the end portion is at an end of the neck portion, and wherein an angle α between an edge of the end portion and the neck portion prevents linear movement of the filter cartridge in an extraction direction opposite an insertion direction parallel to the longitudinal axis.
4. The filter cartridge of claim 3, wherein each end portion is configured such that, during at least a

portion of the insertion of the filter cartridge into the filter cartridge manifold in the insertion direction, at least a portion of each of the first latching mechanism and the second latching mechanism is engaged.

5. The filter cartridge of claim 4, wherein engaging the portion of each of the first latching mechanism and the second latching mechanism comprises causing each of the first latching mechanism and the second latching mechanism to flex radially.

6. The filter cartridge of claim 3, wherein each end portion is structured to be received by a respective channel of the first channel and the second channel and to prevent, after being received by the respective channel, the linear movement of the filter cartridge in the extraction direction without rotation of the filter cartridge.

7. The filter cartridge of claim 1, wherein the first latching mechanism is configured to interact with the curved wall of the first channel to permit, after being received by the first channel, rotation of the filter cartridge in a second direction.

8. The filter cartridge of claim 7, wherein, after insertion of the filter cartridge into the filter cartridge manifold and during the rotation of the filter cartridge in the second direction, the first latching mechanism is structured to interact with the curved wall of the first channel to displace the first latching mechanism to flex radially.

9. A method, comprising: inserting, in an insertion direction, a filter cartridge into a filter cartridge manifold of an appliance, wherein: the filter cartridge comprises: a filter body having a first end, a second end, and a longitudinal axis; an integrated locking head comprising a first component on an outer surface of the integrated locking head; and a key extending radially outward from the filter body adjacent the first end, wherein during insertion of the filter cartridge into the filter cartridge manifold, the key is structured to pass through a primary keyway of the filter cartridge manifold; the filter cartridge manifold comprises the primary keyway and a second component; the first component is configured to engage the second component; one of the first component and the second component is a pair of channels comprising a first channel and a second channel that are diametrically opposed with respect to the longitudinal axis of the filter cartridge, wherein the first channel and the second channel each comprise a curved wall at an end of each respective channel; the other of the first component and the second component is a pair of latching mechanisms comprising a first latching mechanism and a second latching mechanism; the first latching mechanism is configured to be received by the first channel; and the second latching mechanism is configured to be received by the second channel; while inserting, engaging the first latching mechanism and the second latching mechanism; and while inserting and after engaging the first latching mechanism and the second latching mechanism, positioning the first latching mechanism in the first channel and positioning the second latching mechanism in the second channel.

10. The method of claim 9, comprising, while the first latching mechanism is positioned in the first channel and the second latching mechanism is positioned in the second channel: preventing movement of the filter cartridge in an extraction direction opposite the insertion direction; and preventing rotation of the filter cartridge in a first direction.

11. The method of claim 10, comprising, while the first latching mechanism is positioned in the first channel and the second latching mechanism is positioned in the second channel, permitting rotation of the filter cartridge in a second direction opposite the first direction.

12. The method of claim 11, comprising, after positioning the first latching mechanism in the first channel and positioning the second latching mechanism in the second channel, rotating the filter cartridge in the second direction.

13. The method of claim 12, comprising, while rotating the filter cartridge in the second direction, displacing the first latching mechanism and the second latching mechanism to flex radially.

14. The method of claim 13, comprising, after displacing the first latching mechanism and the second latching mechanism to flex radially, pulling the filter cartridge in the extraction direction causing the key to enter a secondary keyway, of the filter cartridge manifold, that intersects with

the primary keyway.

15. A mechanical interlock system, comprising: a filter cartridge comprising: a filter body having a first end, a second end, and a longitudinal axis; a fluid inlet and outlet at the first end of the filter body; an integrated locking head on the first end of the filter body, wherein the integrated locking head comprises a first component on an outer surface of the integrated locking head, wherein the first component is configured to engage a second component of a filter cartridge manifold; and a key extending radially outward from the filter body adjacent the first end; and the filter cartridge manifold comprising a primary keyway and the second component; wherein one of the first component and the second component is a pair of channels comprising a first channel and a second channel, wherein the first channel and the second channel are diametrically opposed with respect to the longitudinal axis, and wherein the first channel and the second channel each comprise a curved wall at an end of each respective channel; wherein the other of the first component and the second component is a pair of latching mechanisms comprising a first latching mechanism and a second latching mechanism, and wherein the first latching mechanism is configured to be received by the first channel and the second latching mechanism is configured to be received by the second channel; and wherein, during insertion of the filter cartridge into the filter cartridge manifold, the key is structured to pass through the primary keyway of the filter cartridge manifold.

16. The mechanical interlock system of claim 15, wherein the first latching mechanism is configured to, after being received by the first channel, sit within the first channel, and wherein the second latching mechanism is configured to, after being received by the second channel, sit within the second channel.

17. The mechanical interlock system of claim 16, wherein, when the key is positioned in the primary keyway of the filter cartridge manifold, the primary keyway is structured to prevent rotation of the filter cartridge until the first latching mechanism is received by the first channel and the second latching mechanism is received by the second channel.

18. The mechanical interlock system of claim 15, wherein each of the first latching mechanism and the second latching mechanism comprises a neck portion and an end portion, wherein each end portion is configured such that, during at least a portion of the insertion of the filter cartridge into the filter cartridge manifold, at least a portion of each of the first latching mechanism and the second latching mechanism is engaged.

19. The mechanical interlock system of claim 15, wherein the first latching mechanism is configured to interact with the curved wall of the first channel to permit, after being received by the first channel, rotation of the filter cartridge in a second direction.

20. The mechanical interlock system of claim 19, wherein, after insertion of the filter cartridge into the filter cartridge manifold and during the rotation of the filter cartridge in the second direction, the first latching mechanism is structured to interact with the curved wall of the first channel to displace the first latching mechanism to flex radially.
